

SANCHARi: A Human-in-the-Loop Geospatial Framework for Satellite Road Network Extraction and Collaborative Curation

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Abstract—Traditional geographic information systems are traditionally built using centralized databases that do not account for rapid landscape changes and natural disasters. This paper proposes a crowdsourced, collaborative GIS with human-in-the-loop collaboration for generating and updating road information using an automated deep learning system, SANCHARI. An advanced U-Net++ architecture SANCHARI is employed that attains approximately 80% Union to segment roads from satellite imagery using an EfficientNet-B4 encoder-decoder architecture. The paper further discusses a collaborative editing framework based on Next.js, LeafletJS, and PostGIS for local administrators to review, edit, lock, and validate spatial vector graphics generated by the deep learning model. The proposed solution is effective in overcoming challenges posed by canopy occlusion and arrival time inaccuracies of automated solutions in diverse terrain and tested across various landscapes in Kerala.

Key Terms - Collaborative GIS, Deep Learning, U-Net++, EfficientNet-B4, OpenStreetMap, WebGIS, Participatory GIS

I. INTRODUCTION

Accurate road and map networks are fundamental geospatial assets required to coordinate disaster response, and on a day-to-day basis, plan for better public administration. However, it becomes a bottleneck to consistently create and maintain such datasets, specifically in regions that frequently undergo dynamic environment changes.

A. The Context of Kerala's Landscape

Kerala's geophysical landscape is particularly challenging for accurate road mapping due to thick vegetation, canal systems, and seasonal flooding. Centralized mapping authorities can be slow to reflect changes on the ground, resulting in inevitably outdated information during emergency situations.

- **Canopy Occlusion:** Dense tropical vegetation and heavy tree plantations consistently tower over village roads and pathways, leading to automated models to completely miss out on detecting such features.
- **Hydrological Complexity:** The extensive networks of inland waterways and canals run parallel to these rural - and often urban - paths, or intersect them, which introduces a whole new layer of spectral ambiguity.
- **Dynamic Vulnerability:** Monsoon-driven floods destroy or temporarily realign crucial last-mile transport links overnight, creating life-and-death stakes for search-and-rescue teams.

II. LITERATURE REVIEW

The research pathway for road extraction as well as map maintenance divulges itself into two areas: The development and evolution of Collaborative WebGIS (online-based systems) as well as the current advancements in Deep Learning and its feature extraction techniques.

A. Collaborative and Web-Based GIS Systems

Collaborative GIS platforms have been improved for distributed urban planning and emergency management. Jing et al. proposed a version-based lightweight collaborative geospatial editing approach for urban planning management that used replication to achieve constraint-free editing. Butt et al. suggested GeoWebEX, an open-source online system for synchronous geospatial collaboration that enables users to share maps, conduct video and audio conferences, and make geo-referenced annotations in a web browser.

Moreover, cloud-based apps were also implemented for local management purposes. Kantartzis and Malesios suggested a decision support system (DSS) for forest road networks management that provided an open-source database of potential problem locations. Netek et al. assessed the use of cloud-based geospatial tools and revealed that GeOnline was the most appropriate such tool for executing process-driven operations directly in browsers. In addition, Pollino et al. employed open-source geospatial tools to design a conceptual model for managing probable damage caused by disastrous earthquakes.

TABLE I
COMPARISON OF COLLABORATIVE GIS AND EXTRACTION FRAMEWORKS

Study / System	Core Contribution	Limitations
Jing et al. (2019)	Developed a version-based lightweight collaborative GIS for urban planning.	Constrained by older, legacy desktop component systems (e.g., Delphi IDE).
GeoWebEX (2018)	Open-source, browser-based synchronous collaboration with geo-referenced annotations.	Relies solely on manual human input; most effective only with GIS-experienced users.
MAiD (2019)	Integrated machine learning into OpenStreetMap editor; increased mapping speeds by 3.5x.	Primarily designed to tackle the addition of major, arterial roads rather than complex rural trails.
Sat2Graph (2020)	Uses Graph-tensor encoding (GTE) to extract network graphs directly, bypassing segmentation limits.	Purely automated framework lacking a human-in-the-loop collaborative editing interface.

B. Machine-Assisted Mapping and Automation

The integration of machine learning into mapping workflows has drastically increased data generation speeds. Bastani et al. created a Machine-assisted iD (MAiD) – an extension to the OpenStreetMap map editor that enables mappers to trace roads using machine assistance. The results show that in a Washington-area experiment, MAiD helped mappers to achieve 3.5x more roads traced than would otherwise be traced manually. Moreover, the number of traced roads exceeded hand-traced roads by 1.7x in a remote region of Indonesia. Qian and Xiaoping engineered a convolutional network based on GANs that reconstructs images with smoother appearance and better defined edges than those produced by standard segmentation procedures.

Recent research has been aimed at improving the topology of the generated results. Thus, Bastani et al. proposed Road-Tracer – an iterative search procedure that utilizes a CNN-based decision function to identify road junctions, yielding to 45% more junctions than standard semantic segmentation approaches. He et al. proposed Sat2Graph, a framework that leverages a novel graph-tensor encoding to directly reconstruct road network graphs from satellite imagery, achieving the best performance across all topological similarity measures.

III. METHODOLOGY

The major novelty of this work resides in its peculiar integration of two separate geospatial paradigms: supervised deep-learning algorithms and participatory GIS (PGIS). Both fully automatic approaches and PGIS are well-established methods for generating standard vector data sets, but each has critical limitations, which, when used in isolation, significantly curtail their effectiveness in practice.

The major shortcoming of fully automatic approaches, implemented with convolutional neural networks, is the inability to deal with the complexities of real-world spatial data structures. In particular, deep-learning models in their current state cannot reliably process occlusions and are generally fairly brittle to variations in environmental conditions, leading to numerous erroneous lines in the generated output graphs, often resulting in fragmented outputs and machine hallucinations (confident, but wrong output from AI models). At the same time, PGIS - the fully manual paradigm of things - suffers from being extremely laborious, if not unrealistically demanding, in terms of the expertise required: only local administrative-level personnel can usually possess the domain knowledge necessary to digitize the vector data correctly.

The innovation of this project was to propose a concept of human-in-the-loop system that combined the strengths of both approaches while minimizing their weaknesses. This was achieved by designing the system around a dual-core design, which consisted of the following two interdependent components:

- 1) **The Automated Bulk Vectorization Engine (SANCHARI):** A high-performance deep-learning backend tasked with the computationally heavy lifting of scanning vast expanses of satellite imagery to generate baseline candidate road geometries instantly.
- 2) **The Collaborative Curation Interface (WebGIS):** A responsive, distributed frontend platform that ingests the machine-generated vectors and empowers local administrative experts to provide critical “last-mile” spatial validation.

These two elements were united in a common database through a shared API, allowing for real-time synchronization of edits and providing the overall coherence of the system. In effect, the proposed system could be used to rapidly generate an initial set of candidate polygons (with roads) and allow authorized local government experts and administrators to edit the automatically generated polygons and lines with their computers while simultaneously viewing changes tracked in real-time by the AI-powered engine. The majority of inauthentic lines were removed by human curators, while the majority of authentic lines were automatically generated by SANCHARI, thus accelerating the process many times over compared to purely manual digitization.

A. Automated Deep Learning Pipeline (SANCHARI Model)

The SANCHARI model is an end-to-end pipeline that accepts a geographic coordinate or a local GeoTIFF file and

returns a georeferenced GeoJSON FeatureCollection of road polylines. The pipeline is illustrated in Fig. 1.

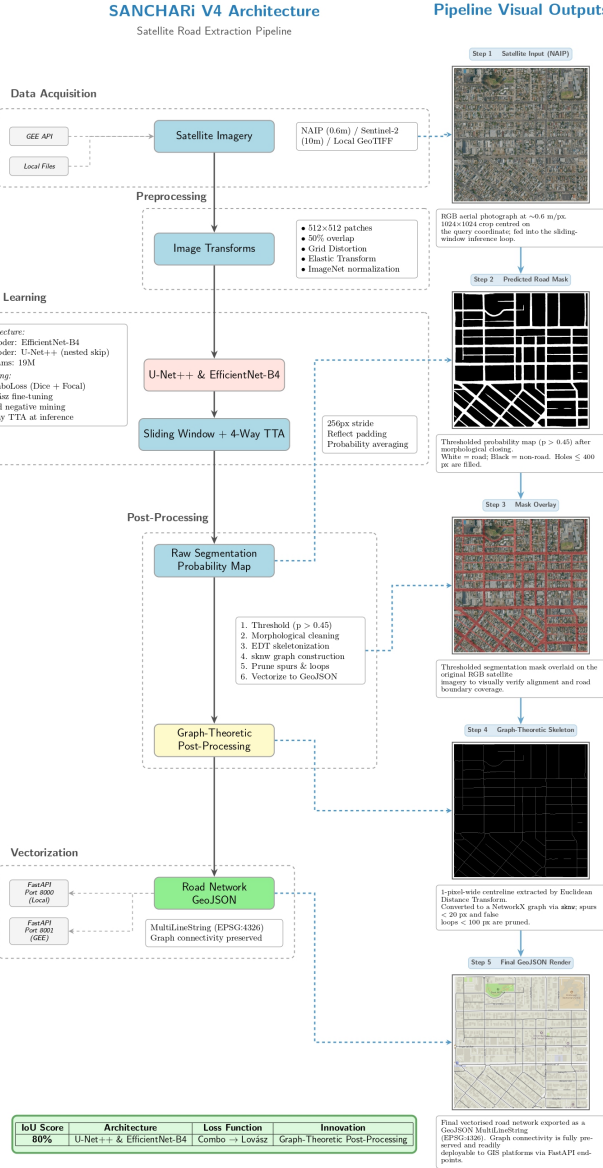


Fig. 1. The end-to-end SANCHARI V4 architecture pipeline, detailing the progression from satellite imagery acquisition to the final vectorisation of the GeoJSON road network [4].

1) *Model Architecture*: The V4 model employs a **U-Net++** decoder with an **EfficientNet-B4** encoder pretrained on ImageNet. U-Net++ replaces the standard single skip connection per depth level with dense nested sub-networks that aggregate features across all encoder scales, closing the semantic gap and preserving thin, single-pixel-wide road centrelines. EfficientNet-B4 uses compound scaling to jointly optimise network depth, width, and resolution, yielding richer texture discrimination than the ResNet34 backbone used in earlier versions.

2) *Training Strategy*: Road extraction is a heavily class-imbalanced binary segmentation task. The V4 model is trained

in three sequential phases.

Phase 1 — Combo Loss. The primary loss combines Dice Loss and Focal Loss in equal proportion:

$$\mathcal{L}_{\text{Combo}} = 0.5 \cdot \mathcal{L}_{\text{Dice}} + 0.5 \cdot \mathcal{L}_{\text{Focal}}$$

Dice Loss optimises pixel-level overlap and handles class imbalance; Focal Loss down-weights easy background pixels and forces the model to focus on hard examples such as road edges obscured by shadows and canopy.

Phase 2 — Hard Negative Mining. After main training, the bottom 20% of training samples by per-image IoU are identified and the model is fine-tuned exclusively on these hard examples, improving performance on deeply occluded intersections and canopy-covered roads.

Phase 3 — Lovász Fine-Tuning. A final fine-tuning phase replaces the loss with the Lovász-Softmax loss [15], which provides a convex surrogation of the Jaccard index and directly optimises the reported IoU metric rather than a differentiable approximation of it.

3) *Inference and Post-Processing*: During inference, input imagery is decomposed into 512×512 patches with 50% overlap. Each patch is processed with **4-way Test Time Augmentation** (original, horizontal flip, vertical flip, 90 rotation), and the four resulting probability maps are averaged before being stitched back into a full-image probability map.

The raw probability map is then refined by a **graph-theoretic post-processing** pipeline rather than simple morphological dilation, which would destroy road topology. The pipeline thresholds the probability map, fills small holes, uses a KD-tree over skeleton endpoints to close small gaps, skeletonises the binary mask to a 1-pixel-wide centreline, and converts the skeleton into a NetworkX graph via `sknw`. Iterative graph pruning then removes short dead-end spurs, false self-loops, and redundant parallel edges. The cleaned graph is finally vectorised and exported as an EPSG:4326 GeoJSON FeatureCollection.

4) *Deployment*: Two Dockerised FastAPI services expose the pipeline: a **GEE API** (port 8001) that fetches live NAIP or Sentinel-2 imagery via the Google Earth Engine API, and a **Local GeoTIFF API** (port 8000) that operates entirely offline against user-supplied `.tif` files.

B. Collaborative Web Architecture

To harness human-enabled context and knowledge about the existing pipeline that the model cannot replicate, this system incorporates a WebGIS architecture to facilitate interactive and collaborative road-network editing, as illustrated in Fig. 2.

- **Frontend Layer & Map Initialization**: The user interface leverages Leaflet.js and Leaflet.draw to support interactive editing of vector data. Upon loading, the map retrieves FlatGeobuf data through HTTP range requests from GeoServer, using a client-side spatial index to speed up data parsing. When a user selects a region, the system uses the area's bounding box to query relevant features, which are then transformed into an in-memory GeoJSON (RFC 7946) copy for editing.

TABLE II
SYSTEM TECHNOLOGY STACK

Layer / Domain	Technologies
Model Architecture	PyTorch, segmentation_models_pytorch, timm
Augmentation	Albumentations (GridDistortion, ElasticTransform)
Spatial Data	rasterio, pyproj, GeoPandas
Graph & Topology	sknw, NetworkX, Shapely, scikit-image
Satellite Imagery	Google Earth Engine API (NAIP / Sentinel-2)
API Services	FastAPI, Uvicorn, Pydantic
Frontend WebGIS	Next.js, React.js, Leaflet.js
Backend & Database	Node.js, Express.js, PostgreSQL/PostGIS, GeoServer
Deployment	Docker, docker-compose (GPU-optional)

- **Client-Side Geometry Processing:** For efficient handling of manual edits, Turf.js acts as the geometry processing engine on the client. Users can either draw LineStrings freehand to add new roads or create polygons to remove existing segments. During deletion operations, functions such as `turf.lineSplit()` and `turf.difference()` are used to split roads at the boundary of the erase area, distinguishing segments to retain from those to remove. All edited geometries are checked for self-intersections before being finalized.
- **Transaction, Sync, and Database:** Synchronization is managed through OGC WFS-T (Web Feature Service – Transactional). Edits made during a session are grouped and formatted into a WFS-T-compliant XML payload, with GeoJSON features converted to GML to represent insertions, updates, and deletions. This payload is sent via HTTP POST to GeoServer, which translates the request into SQL commands executed on the PostGIS database. Changes are applied atomically, and the response—parsed from the returned XML—provides confirmation of the transaction (number of features inserted, updated, deleted) before the updated map view is refreshed in the interface.

IV. RESULTS AND DISCUSSIONS

A. Empirical Case Study Regions

To evaluate operational resilience, validation was conducted across heterogeneous pilot zones in Kerala.

- **Zone A (Melila Panchayat):** Dense multi-tier canopy cover, narrow unpaved tracks, requiring high model sensitivity and local human validation to connect obscured segments.
- **Zone B (Urban Core):** High impervious surface ratios and concrete infrastructures that spectrally mimic road corridors.

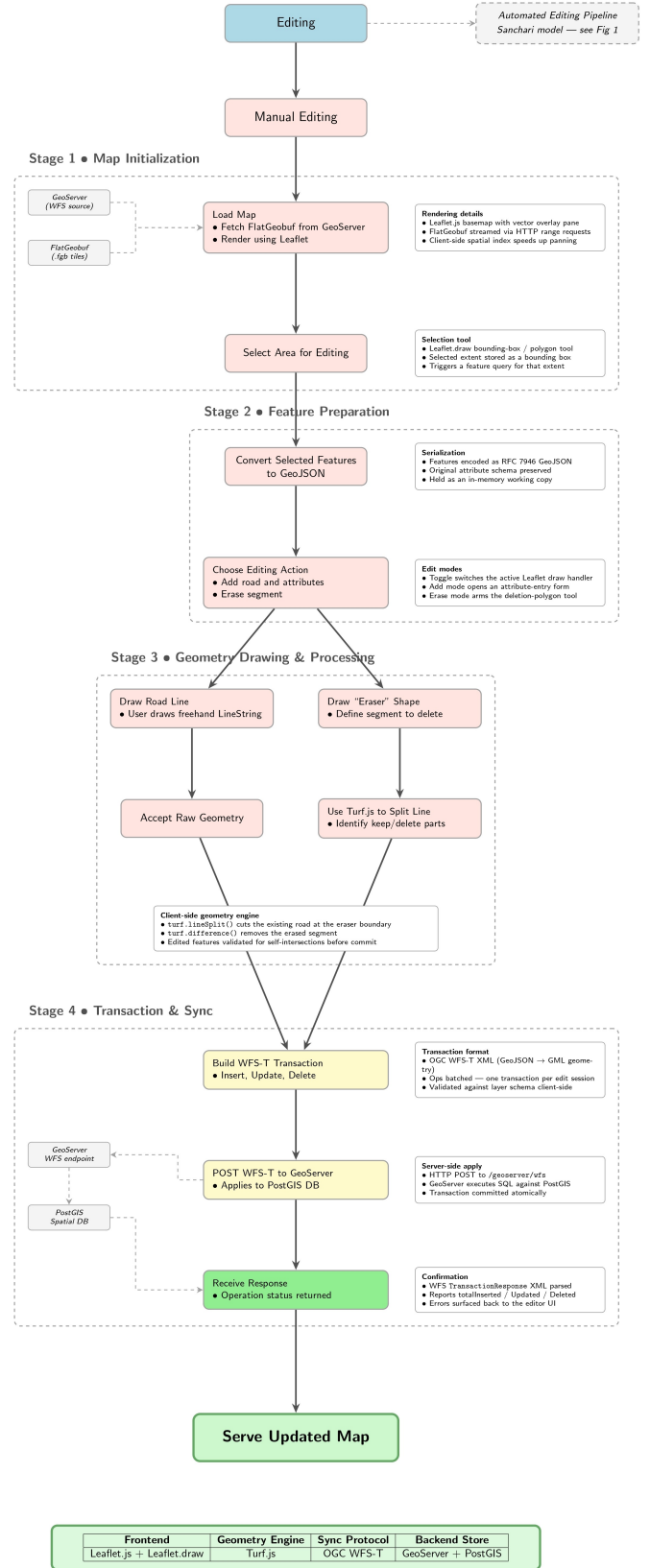


Fig. 2. The SANCHARI Manual Editing Workflow, detailing the four-stage interactive road-network editing pipeline utilizing Leaflet, Turf.js, and GeoServer WFS-T for database synchronization.

- **Zone C (Coastal Region):** High-reflectance sandy soils and roads running parallel to backwaters, testing the model’s spatial context awareness.

B. Model Evolution and Performance

The automated extraction model evolved significantly to address the cartographic demands of the study regions.

TABLE III
MODEL ARCHITECTURE EVOLUTION AND IOU METRICS

Version	Architecture Changes	Validation IoU
V1	Custom U-Net (scratch), BCE Loss	54% - 55%
V2	ResNet34 Encoder, Dice Loss	62.8% - 68%
V3	Sliding Window TTA, Attention Gates	75%
V4	EfficientNet-B4, U-Net++, Combo/Lovász Loss, Graph Pruning	80%

The V1 baseline custom U-Net struggled severely with connectivity, achieving an IoU of 54%. Transitioning to a pre-trained ResNet34 backbone (V2) improved IoU to 62.8%. The final V4 production pipeline, utilizing the EfficientNet-B4 backbone and the graph-theoretic post-processing suite, achieved an IoU of 80%. The inclusion of GridDistortion augmentations and 4-way Test Time Augmentation (TTA) drastically reduced false positives generated by shadows and parking lots in Zone B.

C. System Interface and Usability

Despite achieving 80% automated accuracy, canopy occlusion in Zone A (Melila) consistently introduced disconnects. The collaborative web application provided the necessary intervention.

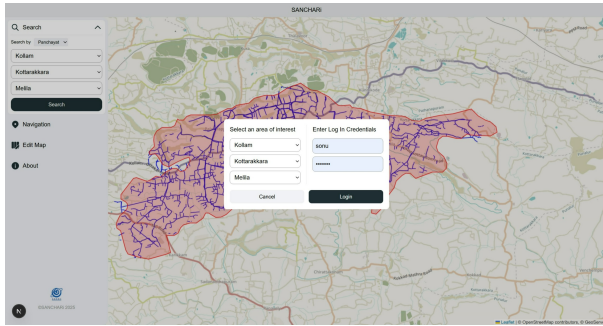


Fig. 3. Home Page Interface locked at Kollam city, demonstrating hierarchical navigation and search tools for local Panchayat administration.

Local administrative staff used the responsive Leaflet web editor to make manual improvements to the map, including connecting dangling road segments, aligning missing intersections, and adding unpaved agricultural paths. The built-in change-log feature maintained a permanent record of all edits, enabling quick reversal of any incorrect or unauthorized changes. This on-the-ground implementation effectively fixed

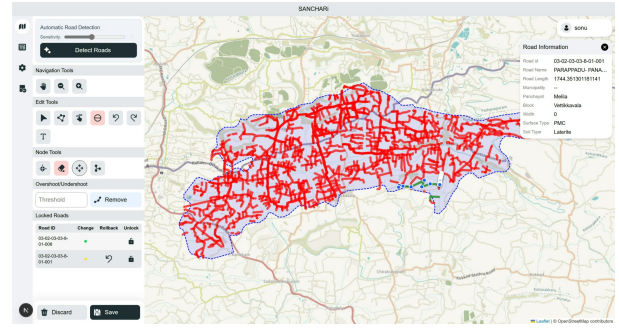


Fig. 4. Sidebar functionalities displaying automated road detection triggers, Leaflet-powered node editing tools, and topological overshoot/undershoot correction limits.

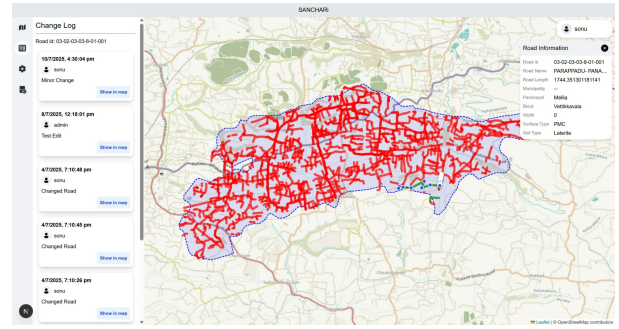


Fig. 5. Highlighting a selected road vector for interactive node modification. The robust PostgreSQL-backed Change Log tracks user edits and enables spatial rollbacks.

long-standing network gaps that had previously disrupted automated route calculations.

V. DISCUSSION AND SOCIO-TECHNICAL IMPACT

The utilization of this collaborative GIS platform for the mapping task has far-reaching implications than just the algorithmic optimization proposed by the authors . The production of authoritative map data has long been the province of either state agencies or commercial entities, resulting in a top-down approach that often rendered local communities unable to contribute their knowledge of the region to formal cartographic records . The resulting data was then subjected to the bureaucracy of the state apparatus with little regard for the immediate and tangible realities of the communities .

The collaborative nature of the system, which uses the SANCHARI deep learning framework combined with a PGIS frontend, has the potential to disrupt the established order . While existing Volunteered Geographic Information (VGI) infrastructures are undeniably more socially just than any centralized alternative, they require significantly higher investments of both time and skill on the part of the volunteer mappers, with only limited rewards to show for the effort . Meanwhile, the fully automated remote sensing paradigm is both inaccessible to non-specialists and unable to capture the nuances of the built environment . The hallucination problem that plagued the early iterations of the system is particularly acute in the forested rural stretches of Kerala,

where the algorithm had difficulty distinguishing between natural vegetative cover and man-made structures such as roads and canal networks .

Meanwhile, the system described in this paper manages to combine the best features of the existing systems while avoiding the pitfalls of extreme automation or extreme decentralization . By utilizing the Efficientnet-B4 network to perform the relatively coarse task of feature detection, we managed to significantly reduce the burden on the end-users while still allowing them to utilize their domain expertise in making the fine adjustments to the generated computer vision model . Local administrators, such as the Panchayat members and municipal survey staff, who may not have had extensive surveying experience, were still able to participate in and benefit from the system . The local knowledge of the region, which is sorely lacking in the fully-automated system, can be utilized by the domain experts to make adjustments to the generated GIS, such as removing roads obscured by the rubber plantations or removing false positives generated by the tile roofing or seasonal flooding .

Furthermore, the practicality of the aforementioned approach is evident, given the context of climate calamities and disasters. The areas under investigation are particularly susceptible to monsoon season flooding and landslides, capable of blocking or disrupting the last-mile connectivity. The process of map updates by centralized authorities creates intolerable delays for such time-sensitive tasks. The developed system, in turn, allows for crowdsourced map triage, responding to the emergency with high efficacy. Specifically, once the catastrophe is over, the AI model can analyze the fresh satellite photos and suggest the updated candidate routes. These routes may then be reviewed, approved, or rejected by local rescue workers and administrators in real-time via the web interface. As a result, the critical infrastructure serving the affected population is guided by the most relevant information, produced by both high-speed algorithms and human expertise.

In conclusion, the described research offers viable insight into the consideration of crowdsourced mapping in extreme conditions. The primary finding is the observation of the current approaches' shortcomings in sustaining effective, up-to-date cartography, especially in the Global South. The operations of local governments are highly reliant on the accuracy of polygon-based digital elevations, which, in turn, requires high expenditures of time and effort. The proposed solution does not seek to undermine the authoritative supervision of existing experts but instead removes the need for laborious vectorization through an automated pipeline. Ultimately, the bridge between computer vision and civic engagement is built, offering a sustainable, replicable framework for continuous topographic audits.

VI. CONCLUSION

This study presents a human-in-the-loop framework that successfully integrates deep-learning-based road extraction with collaborative editing of maps. Upgrading the network backbone to EfficientNet-B4 and optimizing the training

regime with a combination of Lovász-Softmax and Combo Loss enabled achieving 80% IoU for the automated baseline. Pairing it with a responsive front-end, which was implemented in Next.js and Leaflet with PostGIS support, ensures that any errors due to the complexity of the terrain can be quickly corrected by the human-in-the-loop.

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